

Chapter 10

Oligopoly Models and Spatial Competition

“Strategy is about making choices, trade-offs; it’s about deliberately choosing to be different.”

— Michael Porter

Before You Begin: Quick Self-Assessment

1. **What is a Nash equilibrium?** *A strategy profile where no player can increase their payoff by unilaterally changing their strategy.*
2. **What is a dominant strategy?** *A strategy that is the best response regardless of what the opponent does.*
3. **What condition does a monopolist use to choose output?** *$MR = MC$; this equates marginal revenue to marginal cost.*
4. **What is consumer surplus, and how is it measured on a supply-and-demand diagram?** *The area below the demand curve and above the price — a triangle when demand is linear.*
5. **What is the difference between a simultaneous and a sequential game?** *In simultaneous games players choose without observing the opponent; in sequential games at least one player moves after observing the other’s action.*

Connecting Week 9 to Week 10: From 2×2 Games to Oligopoly Markets

Week 9 Established

Nash equilibrium, dominant strategies, the prisoner's dilemma, best-response analysis (circle method). All in the context of abstract 2×2 pay-off matrices.

Left open: How do these tools apply to firms competing in actual markets where players choose prices or quantities?

Week 10 Applies It

Cournot — firms choose quantities simultaneously (Nash in quantities).

Stackelberg — one firm moves first (backward induction → SPNE).

Bertrand — firms choose prices (Nash in prices → paradox).

Hotelling — firms choose locations/characteristics (Nash in positions).

*The recurring theme: **timing matters**. Simultaneous games produce one equilibrium. Sequential games produce a subgame-perfect equilibrium that may differ — first-mover advantage is the key new result.*

Questions This Chapter Will Answer

- Why does a Cournot duopoly produce more than a monopolist but less than a competitive market?
- How does a Stackelberg leader earn \$1,800 while its rival earns only \$900, when both have the same costs?
- Why do two price-setting firms with identical products end up charging marginal cost and earning zero profit?
- What keeps the Bertrand zero-profit result from applying to most real markets?
- Why do petrol stations cluster at the same intersection rather than spacing out along a road?
- Why do political parties converge to the centre rather than maintaining distinct platforms?

Learning Objectives and Topics

Each objective states what you will be able to do.

LO1 Identify dominant strategies; use them to find Nash equilibria without best-response analysis.

§10.2 Dominant Strategies

LO2 Derive Cournot best-response functions; find the Cournot–Nash equilibrium; show output lies between monopoly and competition.

§10.3 Cournot Model

LO3 Solve sequential games by backward induction; identify the SPNE; explain why it eliminates non-credible threats.

§10.4 Sequential Games

LO4 Derive the Stackelberg equilibrium; quantify first-mover advantage; compare to Cournot.

§10.5 Stackelberg Model

LO5 State the Bertrand paradox; identify which assumption(s) break down in specific real markets.

§10.6 Bertrand Model

LO6 Derive the Hotelling location equilibrium; explain the principle of minimum differentiation and its limitations.

§10.7 Hotelling Model

Notation Used in This Chapter.

q_i	firm i 's output	$Q = q_A + q_B$	total output
$P(Q)$	inverse demand	$BR_i(q_j)$	best-response function
π_i	firm i 's profit	SPNE	subgame-perfect Nash equilibrium
a, b	Hotelling locations	x^*	indifferent consumer location
n	number of firms	c	marginal cost

10.1. Introduction

Two airlines both prefer to charge \$400 for a seat. But Boeing and Airbus compete on quantity. Toyota and Walmart choose locations. Why does the strategic variable matter so much for the equilibrium outcome?

Why This Matters. The choice of model is not academic: it changes the predicted price, quantity, and profit. Cournot is appropriate for markets where capacity is fixed in advance (oil, steel). Bertrand fits markets where consumers can switch instantly (retail, online). Stackelberg fits markets where one firm can credibly commit first (Walmart store openings, platform launches). Knowing which model applies is the first step in any competitive analysis.

We begin by sharpening the dominant-strategy concept from Week 9.

10.2. Dominant Strategies

In the prisoner's dilemma (Week 9), both players confess regardless of what the other does. That is a dominant strategy. When does this strong equilibrium concept apply, and what does it imply?

Dominant Strategy: A strategy that yields a strictly higher payoff than any other strategy available to a player, *irrespective of the opponent's choice*. If a dominant strategy exists, a rational player always plays it — no reasoning about the opponent is needed.

In words: A dominant strategy is the best reply no matter what the opponent does. If it exists, playing anything else is irrational.

Two key observations:

- (i) Dominant strategies do not exist in every game.
- (ii) If every player has a dominant strategy, the Nash equilibrium is simply the profile of dominant strategies — no further reasoning required.

*Dominant strategy
 $\Rightarrow$ always part of a
 Nash equilibrium.
 Nash equilibrium
 $\nRightarrow$ involves
 dominant
 strategies.*

10.2.1. Worked Example

Payoffs (A, B) :

		Left	Right
A	Top	(3, 9)	(2, 8)
	Bottom	(0, 2)	(1, 1)

A's best response: if B plays Left, Top ($3 > 0$); if B plays Right, Top ($2 > 1$). **Top dominates for A.**

B's best response: if A plays Top, Left ($9 > 8$); if A plays Bottom, Left ($2 > 1$). **Left dominates for B.**

Nash equilibrium: (Top, Left) with payoffs (3, 9).

Advertising Arms Race. Coca-Cola vs. Pepsi: regardless of whether the rival advertises, each firm earns more by advertising (it steals market share when the rival is silent; it avoids losing share when the rival advertises). Advertising is a dominant strategy for each firm. NE: (Advertise, Advertise), which is Pareto dominated by (Don't, Don't) — a direct parallel to the prisoner's dilemma. This explains persistent, heavy advertising expenditure that earns neither firm a net advantage.

Check Your Understanding

1. In the payoff matrix above, show that (Top, Left) is a Nash equilibrium using the circle method.
2. Consider the new matrix below. Does Firm A have a dominant strategy? Does Firm B? If neither has one, can you still find a Nash equilibrium?

		Left	Right
A	Top	(4, 3)	(1, 4)
	Bottom	(2, 1)	(3, 2)

3. Does a dominant strategy always exist? Give one game where it does (Prisoner's dilemma) and one where it does not.
4. True or false: "If a player's dominant strategy yields a payoff of 2, and the opponent plays a Nash equilibrium strategy, the player earns at least 2."

Dominant strategies settle the equilibrium when they exist. But in oligopoly markets, firms typically choose quantities or prices — we need best-response analysis for the general case.

10.3. The Cournot Model of Oligopoly

OPEC members simultaneously set oil production quotas. Each country treats rivals' output as fixed and maximises its own revenue. What is the equilibrium, and how does it compare to the monopoly and competitive outcomes?

10.3.1. Setup

Market demand: $P = 120 - Q$, $Q = q_A + q_B$. **Marginal cost:** $MC = 0$. Firms choose quantities *simultaneously*.

Cournot Model (1838): Firms simultaneously choose *quantities*, treating rivals' quantities as given. Each firm maximises profit holding q_j constant.

Best-Response Function: Firm i 's profit-maximising quantity as a function of firm j 's quantity. At equilibrium, each firm is simultaneously on its own BRF.

The Cournot Idea in Plain English. Each firm faces an uncomfortable strategic problem: it does not know how much the rival will produce when it makes its own decision. The Cournot assumption cuts through this by saying each firm *takes the rival's output as given* — that is, it treats the rival's quantity as a fixed number and asks: "Given what my rival produces, how much should I produce to maximise my own profit?" The answer to this question is the firm's **best-response function** (also called its reaction function). At the Cournot–Nash equilibrium, both firms are simultaneously best-responding to each other: neither firm wants to change its output given what the other is doing.

10.3.2. Deriving Best-Response Functions

Firm A maximises $\pi_A = (120 - q_A - q_B)q_A$, treating q_B as fixed. Here is the derivation step by step:

Step 1. Write total revenue: $TR_A = P \cdot q_A = (120 - q_A - q_B) \cdot q_A = 120q_A - q_A^2 - q_Bq_A$.

Step 2. Differentiate with respect to q_A , holding q_B constant:

$$MR_A = \frac{dTR_A}{dq_A} = 120 - 2q_A - q_B.$$

Step 3. Set $MR_A = MC = 0$ and solve:

$$120 - 2q_A - q_B = 0 \implies 2q_A = 120 - q_B \implies q_A^*(q_B) = \frac{120 - q_B}{2}.$$

This is Firm A's best-response (reaction) function. (10.1)

This is **Firm A's reaction function**: for every possible value of q_B , it gives Firm A's profit-maximising response. If Firm B produces nothing ($q_B = 0$), Firm A acts as a monopolist and produces 60. If Firm B produces 60, Firm A optimally produces only 30. More rival output $\Rightarrow$ lower optimal own output.

By exactly the same algebra applied to Firm B:

$$q_B^*(q_A) = \frac{120 - q_A}{2}. \quad (10.2)$$

In words: The more Firm B produces, the less Firm A wants to produce. Quantities are strategic substitutes.

Firm A's best-response function is *decreasing* in q_B — a higher rival output depresses market price, reducing the marginal profitability of each additional unit A might produce. This **strate-**

Strategic substitutes: higher q_B lowers A's optimal q_A . Contrast with prices, which are strategic complements.

gic substitutability of quantities is the defining feature of Cournot competition.

10.3.3. The Cournot–Nash Equilibrium

At the Nash equilibrium, both equations hold simultaneously. By symmetry, $q_A^* = q_B^* \equiv q^*$:

$$q^* = \frac{120 - q^*}{2} \implies q^* = 40. \quad (10.3)$$

Cournot–Nash Equilibrium:

$$q_A^* = q_B^* = 40, \quad Q^* = 80, \quad P^* = \$40, \quad \pi_A^* = \pi_B^* = \$1,600.$$

Output lies strictly between the monopoly output (60) and the competitive output (120).

In words: Each firm produces 40 — more than the collusive 30, less than the competitive 60. Neither can profitably deviate.

Why is (40, 40) a Nash equilibrium? Because if Firm A produces 40 and Firm B produces 40, neither firm can increase its profit by unilaterally changing its output. To see this concretely: if Firm A tried to produce 50 instead, total output would rise to 90, price would fall to $P = 120 - 90 = \$30$, and Firm A's profit would be $30 \times 50 = \$1,500$ — less than the \$1,600 it earns at $q_A = 40$. The same calculation applies to Firm B. The Cournot equilibrium is *self-enforcing*: given what the other does, no firm can do better by deviating.

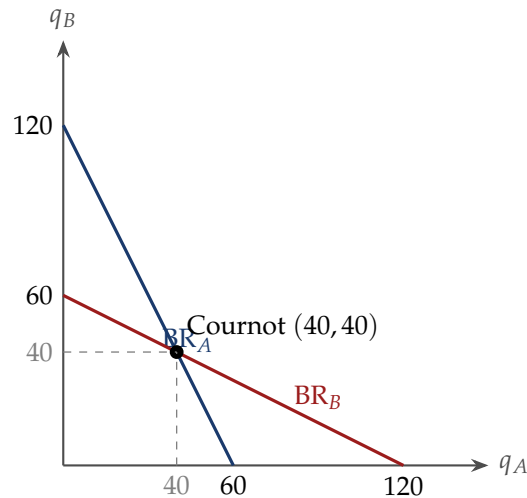


Figure 10.1. Cournot best-response functions. What to look for: BR_A (blue) runs from $(q_A, q_B) = (0, 120)$ to $(60, 0)$. BR_B (red) runs from $(0, 60)$ to $(120, 0)$... wait — BR_B passes through $(0, 60)$ and $(120, 0)$. Their intersection at $(40, 40)$ is the Nash equilibrium: each firm is simultaneously best-responding to the other. If either firm deviates, its profit falls.

10.3.4. Economic Interpretation: The n -Firm Result

Cournot with n Identical Firms ($P = 120 - Q$, $MC = 0$):

$$q_i^* = \frac{120}{n+1}, \quad Q^* = \frac{120n}{n+1}, \quad P^* = \frac{120}{n+1}.$$

n	Per-firm output q^*	Total output Q^*	Price P^*	Outcome
1 (monopoly)	60	60	\$60	Max market power
2 (duopoly)	40	80	\$40	← Chapter benchmark
3 (triopoly)	30	90	\$30	
∞ (competition)	$\rightarrow 0$	$\rightarrow 120$	$\rightarrow \$0 = MC$	Competitive

As $n \rightarrow \infty$, the Cournot price converges to marginal cost. More firms $\Rightarrow$ more competitive outcome.

In plain English:
More Cournot firms = lower price, higher output, less profit each. One firm = monopoly. Infinite = perfect competition.

Check Your Understanding

1. Verify the Cournot NE: at $(q_A, q_B) = (40, 40)$, confirm A's profit = \$1,600 and that deviating to $q_A = 50$ yields only \$1,500. Why does this confirm $(40, 40)$ is a Nash equilibrium?
2. According to Firm B's reaction function $q_B^* = (120 - q_A)/2$, what is Firm B's best response if Firm A suddenly produces 60 instead of 40?
3. How does total Cournot output (80) compare to the monopoly output (60) and the competitive output (120)? Why do both firms produce *more* than the monopoly quantity?
4. True or false: "The Cournot equilibrium is efficient (no deadweight loss)." Explain.

Application: Airline Route Competition as Cournot.

American Airlines and United Airlines both serve the Chicago–Dallas route. Each airline makes capacity decisions (number of seats, flight frequency) months in advance, before observing the rival's schedule for the same period. Neither can instantly adjust once aircraft are committed to routes. This describes the Cournot setup almost exactly: capacity is set simultaneously, treating the rival's capacity as given.

The result matches Cournot predictions: fares on routes dominated by two carriers (a "duopoly route") are substantially higher than on routes with three or more major carriers — a pattern consistently documented in airline economics. When Southwest Airlines enters a duopoly route, fares often fall sharply — consistent with n moving from 2 toward 3 in the Cournot n -firm formula, reducing P^* from \$40 toward \$30.

Similarly, OPEC members setting oil production quotas is classic Cournot: each country takes others' output as given when maximising its own oil revenue.

In Cournot, both firms choose simultaneously. But what if one firm can commit to a quantity before the other decides?

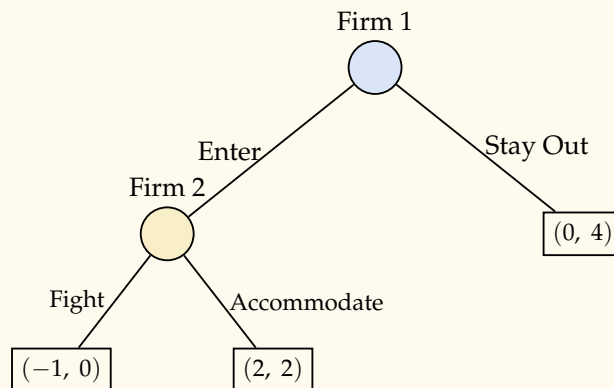
10.4. Simultaneous Versus Sequential Games

The simultaneous game in Section 10.2 had two Nash equilibria: (Top, Left) and (Bottom, Right). If Player A moves first, backward induction selects a unique outcome. Which NE survives, and why?

10.4.1. A Warm-Up: The Entry Game

Before applying backward induction to quantity competition, consider a simple two-stage game that illustrates the method clearly.

The Entry Game. An entrant (Firm 1) decides whether to *Enter* or *Stay Out* of a market. If Firm 1 enters, the incumbent (Firm 2) decides whether to *Fight* (price war) or *Accommodate* (share the market peacefully). Payoffs (Firm 1, Firm 2):



Backward induction — Step 1 (Firm 2's decision if Firm 1 enters):

Fight yields Firm 2 a payoff of 0; Accommodate yields 2. Firm 2 prefers **Accommodate**. Fighting is a *non-credible threat* — once Firm 1 is actually in the market, Firm 2 is better off coexisting.

Backward induction — Step 2 (Firm 1's decision):

Firm 1 anticipates Accommodate, earning 2 if it enters vs. 0 if it stays out. Firm 1 **Enters**.

SPNE: (Enter, Accommodate) with payoffs (2, 2).

The incumbent's implicit threat — "I will fight if you enter" — is not credible because fighting hurts the incumbent once entry has occurred. Backward induction exposes this, selecting the unique SPNE.

The same backward-induction logic applies when there are multiple Nash equilibria in the simultaneous game and we need a principled way to select one.

10.4.2. Multiple Nash Equilibria and the Selection Problem

Many games have multiple Nash equilibria. Consider:

		Left	Right
A	Top	(3, 9)	(1, 8)
	Bottom	(0, 0)	(2, 1)

Best-response check: (Top, Left) and (Bottom, Right) are both Nash equilibria. Additional

The entry-game SPNE explains why mere verbal threats to "fight any entrant" are insufficient. Credibility requires a pre-commitment (e.g. building excess capacity) that makes fighting rational after entry.

reasoning is needed to select between them. A powerful selection criterion: ask which NE survives when the game is played sequentially.

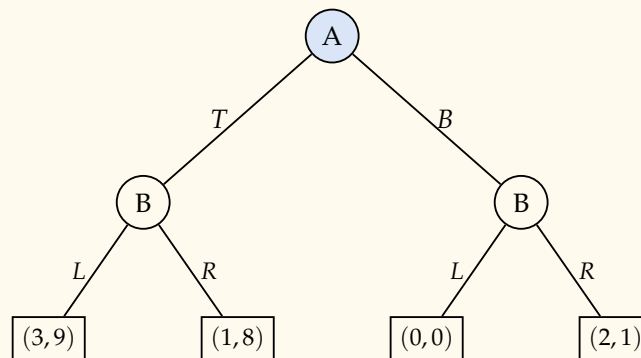
10.4.3. Backward Induction and the SPNE

Backward Induction: Solve a sequential game by starting at the terminal nodes and working backward, at each node choosing the action that maximises the current player's payoff.

Subgame-Perfect Nash Equilibrium (SPNE): A Nash equilibrium that prescribes optimal play at *every* subgame — it eliminates NE sustained by non-credible threats.

In words: Backward induction: reason from the end of the game tree to the beginning, pruning suboptimal branches. The SPNE is the unique outcome of this process.

Backward Induction: A moves first.



Step 1: If A plays Top, B chooses L (payoff $9 > 8$). A gets 3.

Step 2: If A plays Bottom, B chooses R (payoff $1 > 0$). A gets 2.

Step 3: A compares 3 (Top) vs. 2 (Bottom). A chooses **Top**.

SPNE: (Top, Left) with payoffs (3, 9). The simultaneous-game NE (Bottom, Right) is eliminated because B's threat to play L if A plays Bottom is non-credible — once A plays Bottom, B prefers R.

"Every Nash equilibrium of the simultaneous game is also an SPNE." This is false. (Bottom, Right) is a Nash equilibrium of the simultaneous game but is not an SPNE of the sequential game. Backward induction eliminates NE supported by non-credible threats — this is precisely its value.

Check Your Understanding

1. In the backward-induction example: if A plays Bottom, why is B's threat to play L non-credible?
2. A sequential game has three stages. How does backward induction proceed?
3. True or false: "The SPNE always selects the Nash equilibrium that is best for the first mover."

The Stackelberg model applies backward induction directly to quantity competition, producing the

most important result in this chapter.

10.5. The Stackelberg Model

Walmart announces store openings years in advance, publicly committing to serve a large share of local demand. Small retailers, anticipating this, scale back or exit. This is the Stackelberg logic — how does commitment translate into profit?

10.5.1. Setup and the Key Insight

Same market as Cournot ($P = 120 - Q$, $MC = 0$), but now Firm A (the leader) commits to a quantity *first*, and Firm B (the follower) observes and responds.

Commitment is power. The Stackelberg leader produces more than its Cournot quantity because it knows the follower will best-respond by producing less. The leader exploits predictability to claim a larger market share. First-mover advantage arises from the ability to commit credibly and irreversibly.

10.5.2. Step 1 — Follower's Problem

Firm B observes q_A and maximises $\pi_B = (120 - q_A - q_B)q_B$:

$$q_B^*(q_A) = \frac{120 - q_A}{2}. \quad (10.4)$$

Identical to the Cournot BRF. The difference: Firm A *knows this function* before choosing q_A .

10.5.3. Step 2 — Leader's Problem

Firm A substitutes (10.4) into its own profit:

$$\pi_A = \left(120 - q_A - \frac{120 - q_A}{2}\right) q_A = \left(60 - \frac{q_A}{2}\right) q_A.$$

$$\text{First-order condition: } 60 - q_A = 0 \Rightarrow \boxed{q_A^* = 60}. \quad (10.5)$$

The leader produces 60 — 50% more than the Cournot quantity of 40.

10.5.4. Stackelberg Equilibrium Outcomes

$$q_B^* = (120 - 60)/2 = 30.$$

Stackelberg Equilibrium:

$$q_A^* = 60, \quad q_B^* = 30, \quad Q^* = 90, \quad P^* = \$30, \quad \pi_A^* = \$1,800, \quad \pi_B^* = \$900.$$

10.5.5. Comparing Cournot and Stackelberg

Model	q_A	q_B	Q	P	π_A	π_B
Cournot (simultaneous)	40	40	80	\$40	\$1,600	\$1,600
Stackelberg (A leads)	60	30	90	\$30	\$1,800	\$900
Collusion (cartel)	30	30	60	\$60	\$1,800	\$1,800
Bertrand (prices)	60	60	120	\$0	\$0	\$0

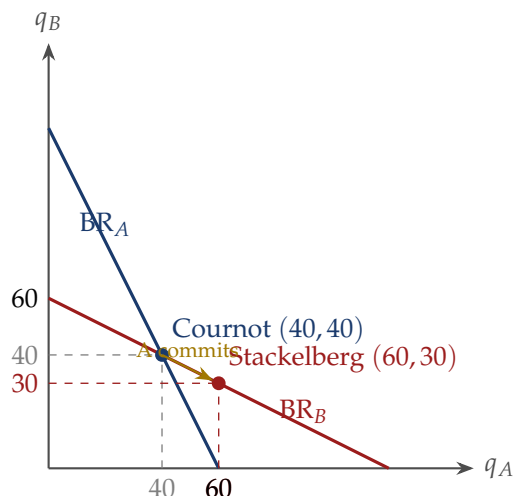


Figure 10.2. Cournot and Stackelberg on the best-response diagram. *What to look for:* The Stackelberg equilibrium (red square) lies on Firm B's reaction curve — B still best-responds — but Firm A has moved from 40 to 60, exploiting its first-mover position. The arrow shows how A's commitment moves the equilibrium along BR_B .

Why Does First-Mover Advantage Exist? The Stackelberg leader commits to a large output level (60 units). The follower, seeing this, knows that also producing a lot would crash the price severely, so it optimally cuts back to 30. The leader *exploits the follower's predictable reaction function*: by choosing 60 rather than 40, the leader moves the equilibrium along BR_B to a point where the follower produces less. This raises the leader's market share and profit.

In numbers: if the leader chose the Cournot quantity (40), the follower would produce 40 and the leader earns $40 \times \$40 = \$1,600$. By committing to 60, the leader induces the follower to produce only 30, giving a price of \$30 and leader profit of $60 \times \$30 = \$1,800$. The \$200 premium is the pure value of moving first.

Connection to business strategy: This is precisely why incumbent firms race to build capacity, sign long-term supply contracts, or lock in distribution partnerships — *commitment confers an advantage*. A capacity investment that would be unprofitable in isolation becomes profitable when it credibly deters rivals from expanding. The key requirement is that the commitment be both *observable* (rivals can see it) and *irreversible* (the committing firm cannot cheaply undo it).

Economic Insight. Four patterns: (i) Leader produces more and earns more than in Cournot. (ii) Follower produces less and earns less. (iii) Total output rises, price falls — consumers

benefit. (iv) Total industry profit falls (\$2,700 vs. \$3,200 in Cournot). The leader's commitment benefits consumers at the industry's expense.

Why First-Mover Advantage Matters for Business. The Stackelberg model formalises one of the most important patterns in competitive strategy: firms that commit early and credibly earn higher profits than rivals who wait.

Amazon's fulfilment network: by building warehouse capacity across the US before rivals could respond, Amazon committed to fast delivery at a scale that made matching the investment prohibitively costly for new entrants.

The iPhone launch (2007): Apple entered the smartphone market with a fully integrated ecosystem (hardware, OS, App Store) before rivals had mobile-optimised platforms. The commitment was observable (announced publicly) and largely irreversible (billions in R&D). Android was forced into a follower position, ultimately settling for a lower-margin strategy (licensing the OS free).

Walmart's supply chain: Walmart built distribution centres and supplier relationships in rural markets where rivals saw no profit. Once established, rivals found the investment to match Walmart's logistics prohibitively expensive, leaving Walmart as the effective monopolist in many markets.

In each case, the logic is Stackelberg's: commit credibly to a large "output" (capacity, ecosystem, network) before the rival decides, exploiting the rival's predictable best-response. The premium over simultaneous (Cournot-like) competition is the value of being first.

Check Your Understanding

1. The leader earns \$1,800 in Stackelberg vs. \$1,600 in Cournot. Show this using the profit formula: $\pi_A = (120 - q_A - q_B^*(q_A)) \cdot q_A$ with $q_A = 60$.
2. The follower earns \$900 in Stackelberg vs. \$1,600 in Cournot. Is the follower better or worse off? Show with numbers. How does it compare to Bertrand (\$0)?
3. If both firms tried to be the Stackelberg leader simultaneously — both choosing 60 — what would the outcome be (price, total output, per-firm profit)? Why is this not an equilibrium?
4. Under what conditions might a firm actually *prefer* to be the follower? Give a real example where waiting is strategically valuable.

The Cournot and Stackelberg models assume firms compete on quantity. What if they compete on price instead?

10.6. The Bertrand Model of Price Competition

Two identical petrol stations at the same intersection, both with cost \$1.20/litre. Both want to charge \$2.00. But each knows it can steal the entire market by charging \$1.99. Where does this logic end?

Bertrand Equilibrium (Paradox): With identical products, unlimited capacity, and simultaneous price-setting, the unique Nash equilibrium is $P_A^* = P_B^* = MC$, with each firm earning **zero profit**.

Two firms suffice to produce the competitive outcome.

In words: If both firms price above MC, either can capture the entire market by cutting price by one cent. This logic drives prices all the way to MC.

Proof sketch:

- If $P_A = P_B > c$: either firm can gain by undercutting by one cent. Not an NE.
- If prices differ with $\min(P_i) > c$: the higher-priced firm sells nothing and can undercut. Not an NE.
- If $P_A = P_B = c$: zero profit; no firm can improve by raising price (loses all customers) or cutting (makes losses). **This is the unique NE.**

10.6.1. When Does the Bertrand Paradox Break Down?

The Bertrand result ($P = MC$, zero profit) depends on three assumptions that frequently fail in real markets. Understanding each failure is more useful than memorising the paradox itself.

1. Capacity constraints. The Bertrand logic requires that a firm cutting price by one cent can serve the entire market. If firms have fixed capacity (e.g. an airline with 150 seats), undercutting does not steal all sales — the rival's seats are still filled before excess demand can be diverted. With binding capacity, the race to the bottom stops before $P = MC$, and equilibrium prices resemble Cournot outcomes. *Example:* Two airlines competing on Chicago–Dallas each operate a fixed number of flights per day; capacity is set months in advance. Neither can serve the entire route demand even if it prices at MC.

2. Product differentiation. If consumers do not view the products as perfect substitutes, a firm can raise its price slightly without losing all customers. Even if Firm B charges \$1.99 and Firm A charges \$2.00, customers who prefer Firm A's location, brand, or service quality may still choose Firm A. The result is a differentiated Bertrand equilibrium with prices above MC and positive profits for both. *Example:* iPhone ($P > \$1,000$) and Samsung Galaxy ($P \approx \800) both price well above marginal cost because switching involves real costs (apps, ecosystem, contacts). Neither captures 100% of the market even when priced higher.

3. Repeated interaction. The Bertrand paradox is a one-shot result. In a market where firms interact repeatedly (every week, every quarter), each firm knows that defecting from a tacit understanding today invites retaliation in all future periods. If the future matters enough (firms are patient), the threat of a price war can sustain prices above MC indefinitely — a result formalised in the theory of repeated games, covered in the Forward Link below.

“Which Assumption Breaks Down?” — The Bertrand Paradox Decision Guide

Assumption that fails	What happens	Real example
Unlimited capacity	Capacity constraint softens undercutting	Airlines, hotels
Identical products	Differentiation gives pricing power	Apple vs. Samsung <i>The</i>
Equal costs	Low-cost firm earns positive profit	EasyJet vs. Ryanair
One-shot game	Repeated interaction enables tacit collusion	Supermarket chains

Bertrand paradox is a calibration tool: if firms do not price at MC, at least one assumption must break down. Identifying which reveals why the market works as it does.

“Bertrand always applies when firms set prices.” The paradox requires *both* identical products *and* unlimited capacity. Airlines, hotels, and streaming services all set prices yet none faces the zero-profit Bertrand outcome — their products are differentiated and/or capacity is constrained.

In plain English:
Bertrand =
compete on price
with identical
products → price
war to MC. Break
one assumption
→ prices above
MC.

Why This Matters. The Bertrand paradox explains why firms invest so heavily in product differentiation, branding, and switching costs. By making their product slightly distinct, they escape the race to zero profit. This is why Coca-Cola spends billions distinguishing itself from generic cola, why Apple maintains a closed ecosystem, and why banks offer loyalty programmes. The model says: if you cannot differentiate, price competition will erode all profit; if you can, even a little differentiation is worth paying for.

For antitrust regulators, the Bertrand model provides a useful benchmark: *markets with two or more price-setting firms and identical products should be highly competitive.* When they are not — when a duopoly still charges well above MC — the regulator should look for the assumption that is failing (capacity limits? differentiation? tacit collusion?) and consider whether intervention is warranted.

Check Your Understanding

1. Bertrand model: $MC = \$10$, demand $Q = 500 - 5P$. Find equilibrium price, quantity, and each firm's profit.
2. Why don't real-world duopolists always undercut each other to MC? Give two specific reasons with market examples.
3. A software market has two firms with $MC \approx 0$ (zero marginal cost of an extra download). What model — Cournot or Bertrand — better predicts their behaviour, and why? Does your answer change if the products are differentiated (e.g. Microsoft Word vs. Google Docs)?
4. Differentiated Bertrand: $q_A = 60 - 2P_A + P_B$, $MC = 0$. Derive Firm A's best-response pricing function.

Bertrand and Cournot both assume homogeneous products. The Hotelling model asks: what if firms compete on location or product characteristics?

10.7. The Hotelling Model of Spatial Competition

Two ice cream vendors are on a 1-mile beach. If Vendor A is at 0.3 and Vendor B at 0.7, each serves 50% of the beach. But each can steal customers by moving toward the other. Where does this process stop?

Think About This First. Imagine two coffee shops deciding where to locate along a 1-kilometre high street. Customers are evenly spread along the street and prefer to buy from the closer shop. Both shops sell identical coffee at the same price. Where should each coffee shop locate to maximise its number of customers?

Think carefully before reading on. Your intuition might suggest they should spread out — locate at the quarter-points (0.25 and 0.75) so that each serves exactly half the street with minimal travel for customers. That arrangement is also socially efficient: total customer travel distance is minimised. But is it a Nash equilibrium?

If Coffee Shop A is at 0.25 and Coffee Shop B is at 0.75, then A serves customers from 0 to 0.50, and B serves 0.50 to 1. Now ask: can A do better by moving? If A moves right to 0.35, the new midpoint becomes $(0.35 + 0.75)/2 = 0.55$, and A now serves customers from 0 to 0.55 — 55% of the street instead of 50%. Moving toward the centre increases A's market share. Coffee Shop B faces the same incentive. Both shops will keep moving toward the centre until they are side by side in the middle — the **principle of minimum differentiation**.

10.7.1. Setup: The Beach Problem

Setup: Beach of length 1 mile. Consumers uniformly distributed. Two shack operators simultaneously choose locations a and $b \in [0, 1]$ with $a < b$. Each consumer patronises the nearest shack. Each operator maximises market share.

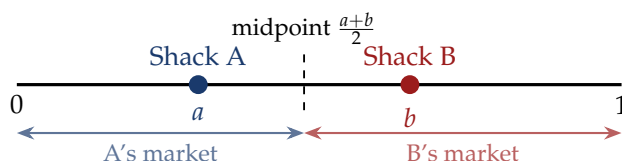


Figure 10.3. Hotelling beach. Consumers left of the midpoint visit Shack A; those right visit Shack B. A's market share $= (a + b)/2$; B's $= 1 - (a + b)/2$.

10.7.2. The Incentive to Move Toward the Centre

Shack A serves $(a + b)/2$ of the beach. Moving A right (increasing a) increases its share: $\partial(\text{A's share})/\partial a = 1/2 > 0$. The same logic applies to B moving left. Both firms have an incentive to creep toward the centre.

Principle of Minimum Differentiation: In the standard Hotelling model with uniform consumer distribution, the unique Nash equilibrium has both firms locate at the **centre**: $a^* = b^* = 1/2$. Neither can gain market share by moving.

In words: Both shacks end up at the midpoint, each serving exactly half the beach. This explains why petrol stations cluster, why political parties converge, and why smartphones look alike.

Application: Platform Competition as Hotelling.

Apple and Google's Android both target the median smartphone user — each wants to maximise the number of users whose preferences are closer to its platform than the rival's. This drives platform *feature convergence*: both platforms now offer similar cameras, app stores, and voice assistants.

However, the Hotelling minimum-differentiation result can break down when price competition is introduced. Once both platforms are at the same location, a slight price cut steals all margin-sensitive users. This creates an incentive to *differentiate*: by being positioned slightly apart, each platform maintains a loyal base that will not immediately switch for a small price difference.

In practice, Apple and Google maintain real differences (open vs. closed ecosystem, premium vs. affordable pricing tiers) partly to soften this price competition — a deliberate departure from minimum differentiation. This is consistent with the extended Hotelling model that incorporates endogenous price-setting alongside location choice.

Application: Minimum Differentiation Across Markets.

The Hotelling equilibrium — both firms at the centre — appears across remarkably different markets:

Smartphones: iOS and Android have converged on nearly identical feature sets over the past decade (similar cameras, voice assistants, app stores, payment systems). Each platform is trying to appeal to the median user. The minimum-differentiation logic is in full operation: neither can afford to be “too different” and thereby lose the mass-market customer.

Political parties: In two-party systems, parties converge toward the median voter's preferences. A party positioned far from the centre concedes the middle — the most votes — to the opponent. The median voter theorem (the political-economy analogue of Hotelling minimum differentiation) is one of the most empirically robust results in political science.

Fast food clustering: McDonald's and Burger King locate near each other (often across the street). Both chains want to capture the customer who has already decided “I want fast food” but not yet which brand. By clustering, they compete for this customer; spreading out would give each a local monopoly but cede the *entire* demand at the rival's location.

Why Minimum Differentiation Matters for Business Strategy. The Hotelling result has a counterintuitive implication: *competing products often look similar not because firms copy each other carelessly, but because it is strategically optimal*. If your product is positioned far from a rival, you serve your loyal customers but surrender the swing customers in the middle. Moving toward the rival steals those swing customers.

This explains: why luxury sedans (BMW 3-Series, Audi A4, Mercedes C-Class) have converged on virtually the same specifications; why major airline service tiers (business, premium economy, economy) are nearly identical across carriers; why major banks offer the same basic current account, mortgage, and savings products. In each case, the Hotelling incentive is at work: differentiate too much and you cede the median customer to a competitor who is willing to position closer to that customer.

The key *limit* on this logic: product differentiation can be valuable precisely when it softens

price competition (the Bertrand paradox resolution). Firms must balance “steal customers by moving closer” against “maintain pricing power by staying distinct.”

The Hotelling result — both firms at the centre — is compelling but not universal. The extensions below show when it breaks down.

10.7.3. Extensions and Limitations

The basic model abstracts from price competition. When price-setting is introduced alongside location choice, firms have an incentive to *differentiate* their positions to soften price rivalry — a force working *against* minimum differentiation. The equilibrium location depends on the relative strength of the “steal customers by getting closer” motive versus the “reduce price competition by staying far” motive.

Median voter theorem (political economy): Political parties converging toward the median voter is the direct political-economy analogue. In practice, parties maintain distinct platforms because: (i) core voters may abstain if the party moves too far from their preferences; (ii) parties can differentiate on non-policy dimensions (candidate quality, competence) even while converging on policy.

Check Your Understanding

1. A is at 0.2, B at 0.7. Compute each firm’s market share. If A can relocate, where does it move to maximise share?
2. Why does introducing price competition undermine the minimum differentiation result?
3. Give one example of firm clustering and one of deliberate differentiation. Explain the market force driving each.

With all four models covered, we close with the chapter review materials.

Welfare Comparison Across Market Structures ($P = 120 - Q$, $MC = 0$)

Structure	Q^*	P^*	CS	PS/Profit	TS	DWL
Monopoly	60	\$60	\$1,800	\$3,600	\$5,400	\$1,800
Cournot (duopoly)	80	\$40	\$3,200	\$3,200	\$6,400	\$800
Stackelberg	90	\$30	\$4,050	\$2,700	\$6,750	\$450
Bertrand (= PC)	120	\$0	\$7,200	\$0	\$7,200	\$0

Rankings: TS: Bertrand > Stackelberg > Cournot > Monopoly. CS: same ranking (consumers prefer competition). PS: Monopoly > Cournot > Stackelberg > Bertrand. Firms prefer monopoly; consumers prefer Bertrand.

Intuition: The more competitive the structure, the closer Q^ is to the efficient level (120 when $MC = 0$), and the smaller the DWL. But more competition shifts surplus from producers to consumers.*

Reading the Welfare Table: Three Lessons.

Why Stackelberg output exceeds Cournot output. The Stackelberg leader exploits the follower's reaction function, committing to 60 units. The follower best-responds with only 30. Total output is 90 — above the Cournot 80. The leader's strategic commitment drives total production beyond what two simultaneous competitors would choose, pushing price down and benefiting consumers.

Why Bertrand output equals the competitive output. Bertrand price competition is so intense that even with just two firms, the equilibrium price is driven to MC. No firm has any market power. The result is efficient (no DWL) but unsustainable (zero profits attract no long-run entry incentive). Bertrand is the only structure in which $P = MC$ with multiple firms still present.

Antitrust implications. Regulators measure welfare using total surplus (or consumer surplus). The table shows that more competitive structures are better for society and for consumers, but worse for producers. This explains why antitrust enforcement targets mergers that move markets from Cournot-like (multiple firms) toward monopoly: each step up in concentration reduces total output, raises price, and creates deadweight loss. The Bertrand outcome — achievable in industries with homogeneous products and no capacity constraints — is the ideal benchmark. Markets that fall short of it may warrant scrutiny.

Where Does This Take Us?

- **Repeated games and collusion:** In a one-shot game, collusion is not a Nash equilibrium — each firm defects to the Cournot output. In a repeated game, the threat of future punishment can sustain cooperation. Whether collusion is stable depends on the discount factor and how quickly cheating is detected.
- **Product differentiation:** The Bertrand paradox dissolves with product differentiation. The differentiated Bertrand model (Section 10.6 Q16) shows prices above MC and positive profits — the foundation of modern oligopoly theory.
- **Entry deterrence:** A Stackelberg leader can commit to excess capacity to deter entry. The credibility of the commitment determines whether potential entrants are actually deterred.
- **Auction theory and mechanism design:** Bidding one's true valuation is a dominant strategy in a Vickrey auction — the dominant-strategy concept from Section 10.2 applied to market design.
- **General equilibrium:** The welfare table above quantifies deadweight loss at the industry level. General equilibrium analysis extends this to economy-wide welfare, including input markets and downstream effects.

Six Common Exam Mistakes — and How to Avoid Them

1. **Confusing Stackelberg and Cournot timing.** In *Stackelberg*, firms move *sequentially* — one firm chooses first and commits before the other responds. In *Cournot*, firms choose *simultaneously* — neither observes the other's decision before committing. This distinction is everything: solving Stackelberg by treating q_B as fixed (like Cournot) gives the wrong answer. Always substitute the follower's BRF into the leader's profit before taking the first-order condition.

2. **Forgetting to substitute the follower's BRF into the leader's profit in Stackelberg.** Maximising $\pi_A = (120 - q_A - q_B)q_A$ holding q_B fixed gives the Cournot result — not Stackelberg. Always substitute $q_B^*(q_A) = (120 - q_A)/2$ first, reducing π_A to a function of q_A alone, then take the FOC.
3. **Confusing the Cournot NE with the collusive (cartel) optimum.** Collusion maximises joint profit: each firm produces 30 and earns \$1,800. The Cournot NE has each firm producing 40 and earning \$1,600. Collusion is more profitable per firm but is not a Nash equilibrium — each firm has an incentive to deviate to $q = 45$, earning \$2,025.
4. **Claiming Bertrand always applies when firms set prices.** The Bertrand paradox requires both identical products and unlimited capacity. Airlines, hotels, and differentiated retailers all set prices but face neither constraint fully — prices are above MC in all these markets.
5. **Applying the minimum differentiation result when price competition is present.** The Hotelling clustering result holds when only location is chosen and prices are fixed. With simultaneous price competition, firms want to differentiate *away* from the centre to soften price rivalry. Do not apply minimum differentiation to markets where firms also compete intensely on price.
6. **Treating a dominant strategy equilibrium as Pareto optimal.** The advertising arms race and prisoner's dilemma show that the dominant strategy equilibrium (Advertise, Advertise) is Pareto dominated by (Don't, Don't). A dominant strategy equilibrium is stable, not efficient. Always check whether the NE is Pareto dominated before asserting it is a good outcome.

Chapter at a Glance — Week 10: Oligopoly Models

Your one-page revision guide.

Key Results (benchmark: $P = 120 - Q$, $MC = 0$)

Model	q_A	q_B	P	π_A
Monopoly	60	—	\$60	\$3,600
Collusion	30	30	\$60	\$1,800
Cournot	40	40	\$40	\$1,600
Stackelberg	60	30	\$30	\$1,800
Bertrand	60	60	\$0	\$0

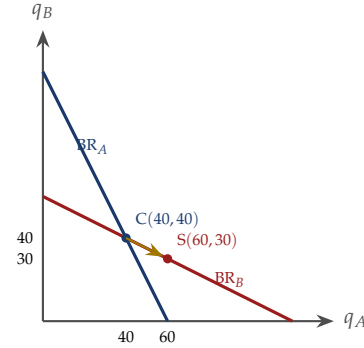
Key Formulas

Cournot BRF	$q_i^* = (120 - q_j)/2$
n -firm Cournot	$q^* = 120/(n+1)$, $P^* = 120/(n+1)$
Stackelberg leader	$q_A^* = (A - c)/2$ (general)
Stackelberg follower	$q_B^* = (A - c)/4$ (general)
Hotelling share (A)	$(a + b)/2$
Hotelling NE	$a^* = b^* = 1/2$

Bertrand Paradox Resolutions

- Capacity constraints $\rightarrow P > MC$
- Product differentiation $\rightarrow P > MC$, $\pi > 0$
- Asymmetric costs $\rightarrow$ lower-cost firm earns $\pi > 0$
- Repeated games $\rightarrow$ tacit collusion at $P > MC$

Best-Response Diagram



LO Checklist

- LO1: Dominant strategy $\rightarrow$ always part of NE ✓
- LO2: Cournot BRF, NE, n -firm convergence ✓
- LO3: Backward induction $\rightarrow$ SPNE ✓
- LO4: Stackelberg: commit $\rightarrow$ first-mover advantage ✓
- LO5: Bertrand paradox + resolutions ✓
- LO6: Hotelling: minimum differentiation + limits ✓

The thread: Dominant strategy (always best regardless of opponent) $\rightarrow$ Cournot: simultaneous quantities, BRFs intersect at (40,40), output between monopoly and competition $\rightarrow$ Backward induction eliminates non-credible threats $\rightarrow$ Stackelberg: leader commits to 60, follower responds with 30, leader earns \$1,800 (\$200 more than Cournot) $\rightarrow$ Bertrand: price competition drives P to MC , zero profit $\rightarrow$ Hotelling: location competition drives both firms to the centre. **Timing and strategic variable together determine the equilibrium.**

10.8. Key Terms

Dominant Strategy

Best response to every opponent strategy; always part of a Nash equilibrium if it exists. *Used throughout the course.*

Cournot Model

Firms simultaneously choose quantities; rivals' quantities treated as given.

Best-Response Function

Firm i 's profit-maximising quantity (or price) as a function of the rival's choice.

Strategic Substitutes

Strategies where an increase in one player's choice reduces the other's optimal choice; Cournot quantities are strategic substitutes.

Cournot–Nash Equilibrium

Intersection of best-response functions; both firms are simultaneously best-responding.

n -Firm Cournot

$q^* = 120/(n + 1)$; price converges to MC as $n \rightarrow \infty$.

Stackelberg Model

Sequential quantity competition; leader commits first, follower responds.

First-Mover Advantage

Benefit from committing to a quantity/action first; requires commitment to be observable and irreversible.

Backward Induction

Solving sequential games from terminal nodes backward; yields the SPNE.

Subgame-Perfect NE (SPNE)

NE that prescribes optimal play at ev-

ery subgame; eliminates NE supported by non-credible threats. *Used throughout the course.*

Credible Commitment

Binding, observable, irreversible action that rivals can rely on; strategically effective only if all three conditions hold.

Bertrand Model

Firms simultaneously set prices for identical products.

Bertrand Paradox

With identical products and unlimited capacity, duopoly competition drives $P = MC$ and zero profit.

Extensive Form

Game-tree representation showing sequence, information, and payoffs.

Hotelling Model

Firms choose locations (or characteristics) on a line; consumers patronise the nearest firm.

Principle of Minimum Differentiation

Hotelling NE: both firms locate at the median consumer preference (the centre).

Median Voter Theorem

Political analogue: competing parties converge to the median voter's preferences.

Strategic Complements

Strategies where an increase in one player's choice raises the other's optimal choice; Bertrand prices are strategic complements.

Recommended Videos

Four videos selected for clarity and direct relevance to this chapter.

1. Cournot Oligopoly (*William Spaniel, Game Theory 101*) ~ 11 min

<https://www.youtube.com/watch?v=EtQiaawrXy0>

Derives Cournot best-response functions, solves for equilibrium quantities and price, and shows how the n -firm case converges to competition. Reinforces Section 10.3 and recommended before Level 2 Cournot questions.

2. Sequential Games and Backward Induction (*Lectures Reuploaded*) ~ 15 min

<https://www.youtube.com/watch?v=RKZDVCDQB2w>

Covers sequential games and backward induction with clear worked examples. Explains the extensive form, the SPNE concept, and why non-credible threats are eliminated. Reinforces Sections 10.4–10.5.

3. Bertrand Oligopoly and Equilibrium (*Justin Eloriaga*) ~ 9 min

<https://www.youtube.com/watch?v=Ru-N3iatJmU>

Explains the zero-profit Bertrand result and the conditions under which prices above MC can arise. Reinforces Section 10.6 and the Bertrand decision guide.

4. The Hotelling Model (*Thomas Nechyba*) ~ 8 min

<https://www.youtube.com/watch?v=QoG0v-Hh17k>

Covers the Hotelling beach setup, minimum differentiation, and the median voter theorem analogue. Reinforces Section 10.7 with clear diagrams.

10.9. Practice Questions

How to use these questions. Difficulty: ★ easy ★★ intermediate ★★★ challenging. Payoffs listed as (Player A, Player B); higher is better unless stated. Attempt every question independently before consulting solutions.

Level 1: Conceptual and Short-Answer

- ★ **[Dominant Strategy — Concept]** Define a dominant strategy. Explain why it must be part of a Nash equilibrium. Can a Nash equilibrium exist without dominant strategies? Give a brief example.
- ★ **[Dominant Strategy — Application]** Payoff matrix (A, B): Up-Left (4, 4); Up-Right (1, 6); Down-Left (6, 1); Down-Right (2, 2).
 - (a) Does A have a dominant strategy? Does B?
 - (b) Find all Nash equilibria.
 - (c) Identify the structure of this game (prisoner's dilemma, coordination, other?).
- ★ **[Cournot — Intuition]** Explain why the Cournot equilibrium output lies between the monopoly output and the competitive output. What happens to the Cournot price as the number of firms increases without bound?
- ★ **[Cournot — Nash Stability]** Cournot: $P = 120 - Q$, $MC = 0$. Firm B produces $q_B = 40$.
 - (a) Firm A's profit at $q_A = 40$.
 - (b) Firm A's profit at $q_A = 50$.
 - (c) Does Firm A want to deviate from the Cournot equilibrium?
- ★ **[Sequential Game — Backward Induction]** A moves first (Top/Bottom); B moves second (Left/Right). Payoffs (A,B): TL=(5, 3), TR=(2, 7), BL=(4, 1), BR=(6, 2).
 - (a) Draw the game tree.
 - (b) Find the SPNE.
 - (c) If played simultaneously, what are the Nash equilibria?
- ★ **[Bertrand — Paradox]** Bertrand: $MC = \$10$, demand $Q = 500 - 5P$.
 - (a) Equilibrium price, quantity, each firm's profit.
 - (b) Monopoly outcome (same demand and cost).
 - (c) Who gains and who loses moving from monopoly to Bertrand duopoly?
- ★ **[Hotelling — Location Game]** Firms on $[0, 1]$, consumers uniformly distributed. Firm A at a , Firm B at $b > a$.

- (a) Each firm's market share.
- (b) Show A has an incentive to move right if $a < 1/2$.
- (c) Nash equilibrium of the location game.

★ **[First-Mover Advantage — Concept]**

- (a) Why does the Stackelberg leader earn more than a Cournot firm?
- (b) Give a real example of a commitment that is both observable and irreversible; one that fails at least one condition.
- (c) The follower earns \$900 in Stackelberg vs. \$1,600 in Cournot. Is the follower harmed?
- (d) When might a firm prefer to be the follower?

★ **[Welfare Across Structures]** $P = 120 - Q$, $MC = 0$.

- (a) Compute CS, PS, TS under: (i) monopoly, (ii) Cournot duopoly, (iii) Stackelberg (A leads), (iv) Bertrand.
- (b) Rank by total welfare. Explain the ranking.
- (c) Rank by consumer surplus. Which do consumers prefer?
- (d) Which structure maximises total welfare? Why is it typically not sustainable?

★ **[Hotelling with Prices]**

- (a) Why might introducing price competition undermine minimum differentiation?
- (b) Describe the trade-off between “stealing customers by moving closer” and “softening price competition by moving further away.”
- (c) Give one real example of clustering and one of deliberate differentiation. Explain each.

★ **[Explain to a Friend — Level 1]** *No formulas.* Your friend says: “In Cournot, each firm treats the rival's output as fixed. But if both firms produce 40 and each knows the other is producing 40, why don't they both agree to produce only 30 and earn more?” Write 3–4 plain-English sentences explaining why collusion at (30,30) is unstable.

Level 2: Intermediate Problems

★★ **[Cournot — Worked Example]** $P = 90 - Q$, $MC = 0$.

- (a) Each firm's BRF.
- (b) Cournot–Nash quantities, price, per-firm profit.
- (c) Sketch the two BRFs and label the equilibrium.

★★ [Cournot — Asymmetric Costs] $P = 100 - Q$, $MC_A = 10$, $MC_B = 20$.

- (a) Each firm's BRF.
- (b) Cournot equilibrium quantities, price, and profits.
- (c) Which firm produces more? Why?

★★ [n -Firm Cournot] $P = 120 - Q$, $MC = 0$, n identical firms.

- (a) Symmetric NE: quantity per firm, total output, price — all as functions of n .
- (b) As $n \rightarrow \infty$, what happens to equilibrium price?
- (c) Complete the table for $n = 1, 2, 3, \infty$.

★★ [Data-Based: Welfare Across Structures] Using $P = 120 - Q$, $MC = 0$, complete the welfare comparison table.

Structure	Q^*	P^*	CS	PS	TS	DWL
Monopoly						
Cournot (duopoly)						
Stackelberg						
Collusion						
Bertrand ($P = MC$)						

- (a) Fill in all cells. Show your calculations.
- (b) Rank by TS. Explain the ranking intuitively.
- (c) Rank by CS. Which do consumers prefer?
- (d) Why does the Stackelberg outcome have lower TS than Bertrand but higher TS than Cournot?
- (e) Collusion achieves the monopoly output but splits it between two firms. Is the joint collusive profit higher than the joint Cournot profit? By how much?

★★ [Stackelberg — Verification and Role Reversal] $P = 120 - Q$, $MC = 0$, Firm A leads.

- (a) Verify: $q_A^* = 60$, $q_B^* = 30$, $P^* = 30$, $\pi_A^* = \$1,800$.
- (b) Now Firm B leads. Find equilibrium quantities, price, and profits.
- (c) Quantify the first-mover advantage. By how much does the leader benefit vs. Cournot?

★★ [Differentiated Bertrand] $q_A = 60 - 2P_A + P_B$, $q_B = 60 - 2P_B + P_A$, $MC = 0$.

- (a) Each firm's best-response pricing function.

(b) Nash equilibrium prices, quantities, profits.

(c) Does the Bertrand paradox apply? Why not?

★★ [Hotelling — Market Shares with Numbers] Beach length 1. A at $a = 0.2$, B at $b = 0.7$.

(a) Each firm's market share.

(b) If A can relocate (holding B fixed at 0.7), where does A move to maximise share?

(c) Describe the adjustment process and final Nash equilibrium.

★★ [Collusion vs. Cournot] $P = 120 - Q$, $MC = 0$.

(a) Collusive output, price, and per-firm profit.

(b) Compare collusive, Cournot, and Bertrand profits per firm.

(c) Show collusion is not a Nash equilibrium: compute profit from deviating to $q_A = 45$ given $q_B = 30$.

★★ [Explain to a Friend — Level 2] Your friend says: "I understand the Bertrand paradox in theory, but I've never seen two firms actually price at marginal cost. The model must be wrong." Write 4–5 sentences explaining which assumptions break down in practice, using two specific market examples.

Level 3: Advanced Problems

★★★★ [General Cournot] $P = a - bQ$, $MC_A = MC_B = c$, $a > c > 0$, $b > 0$.

(a) General BRFs.

(b) General Cournot equilibrium quantities and price.

(c) Show $P^* = (a + 2c)/3$. Verify it converges to c as $a \rightarrow c$.

★★★★ [Asymmetric Stackelberg] $P = 100 - Q$, $MC_A = 0$ (leader), $MC_B = 10$ (follower).

(a) Follower's BRF.

(b) Leader's optimal quantity, price, profit.

(c) Compare to symmetric Cournot ($MC_A = MC_B = 0$). How do asymmetric costs affect outcomes?

★★★★ [General Stackelberg] $P = A - Q$, $MC = c$ for both.

(a) Leader's and follower's equilibrium quantities in terms of A and c .

(b) Show the leader always produces more than the follower.

(c) Compute both profits. Show $\pi_A^* > \pi_B^*$.

(d) Show the leader's Stackelberg profit exceeds its Cournot profit.

- ★★★ [Extensive-Form: Three-Stage Game] Firm A moves first (H or L); Firm B responds (H or L); Firm A moves again (H or L). Terminal payoffs (A, B): HHH(2, 5), HHL(4, 3), HLH(6, 2), HLL(5, 4), LHH(1, 6), LHL(3, 4), LLH(7, 1), LLL(4, 3).

Find the SPNE by backward induction.

- ★★★ [Commitment and Value of Moving First] $P = 120 - Q$, $MC = 0$. Before the Cournot game, Firm A can publicly commit (Stackelberg) or keep plans private (Cournot) at commitment cost F .

- (a) Firm A's profit if it commits.
- (b) Firm A's profit if it does not commit.
- (c) For what values of F does Firm A choose to commit?
- (d) Why is the ability to commit valuable even when it does not change marginal cost?

- ★★★ [Explain to a Friend — Level 3] Your friend, a business student, says: "The Stackelberg model says moving first is always better. So every firm should try to commit first to a large quantity. But that can't be right — otherwise we'd never see Cournot-like simultaneous decisions in practice." Write a 5–6 sentence response addressing: (a) why commitment is not always possible; (b) what makes a commitment credible; (c) why some markets naturally favour simultaneous decisions; and (d) one real market where sequential commitment happens and one where it does not, explaining the difference.

Full solutions follow in the solutions document. Attempt all questions independently before consulting them.